

**Time:** 5 minutes**Outcome:** Explain how pairwise majority rule can generate cyclic collective preferences.**Difficulty:**

Intermediate

Condorcet Paradox & Voting Cycles



THE CORE IDEA

Even when every voter has a consistent ranking, pairwise majority voting can produce a cycle: A defeats B, B defeats C, and C defeats A. The group then has no option that defeats every alternative, and agenda order can affect the final outcome.



HOW TO RECOGNIZE IT

- There are at least three alternatives and voters rank them differently.
- Compare alternatives two at a time using majority rule.
- The collective relation cycles even though each individual's ranking is consistent.
- Because no option is a universal majority winner, agenda order may matter.



WATCH OUT

A voting cycle does not require irrational voters. Individually transitive rankings can aggregate into an intransitive majority relation.



WORKED EXAMPLE: THREE VOTERS, THREE OPTIONS

Individual rankings (first to third)

	First choice	Second choice	Third choice
Voter 1	A	B	C
Voter 2	B	C	A
Voter 3	C	A	B

Using the rankings, A defeats B by 2 votes to 1; B defeats C by 2 to 1; and C defeats A by 2 to 1. Each voter has a consistent ranking, yet the group cycles. With no option beating every rival, the order of pairwise votes can affect the final survivor.



CHECK YOURSELF

Why can changing the order of pairwise votes change the winner when majority preferences form a cycle?

**READY?**

Return to the game and master this concept.